JEE

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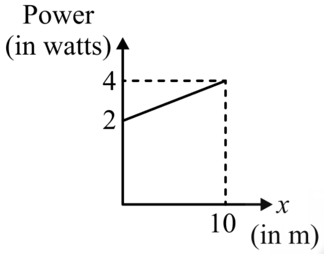
Physics

Work, Energy and Power

DPP: 08

Q1 A particle A of mass $\frac{10}{7}$ kg is moving in the positive x-direction. Its initial position is $x=0$ & initial velocity is 1m/s. The velocity at $x=10$ m is : (use the graph given)



(A) $4 \mathrm{~m/s}$

(B) $2 \mathrm{~m/s}$

(C) $3\sqrt{2} \mathrm{~m/s}$

(D) $100/3 \mathrm{~m/s}$

Q2 A force acts on a $30 \mathrm{~gm}$ particle in such a way that the position of the particle as a function of time is given by $x=3t-4t^2+t^3$, where x is in metres and t is in seconds. The work done during the first 4 seconds is

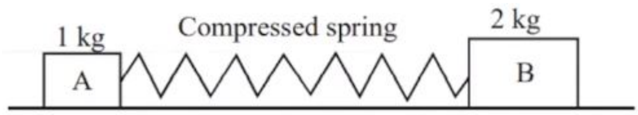
(A) $5.28 \mathrm{~J}$

(B) $450 \mathrm{~mJ}$

(C) $490 \mathrm{~mJ}$

(D) $530 \mathrm{~mJ}$

Q3 An elastic spring is compressed between two blocks of masses $1 \mathrm{~kg}$ and $2 \mathrm{~kg}$ resting on a smooth horizontal table as shown. If the spring has $12 \mathrm{~J}$ of energy and suddenly released, the velocity with which the larger block of $2 \mathrm{~kg}$ moves will be



(A) 2 m/s (B) 4 m/s

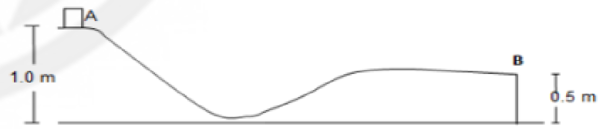
(C) 1 m/s (D) 8 m/s

Q4 When a $1.0 \mathrm{~kg}$ mass hangs attached to a spring of length $50 \mathrm{~cm}$, the spring stretches by $2 \mathrm{~cm}$. The mass is pulled down until the length of the spring becomes $60 \mathrm{~cm}$. What is the amount of elastic energy stored in the spring in this condition, if $g=10 \mathrm{~m/s^2}$

(A) 1.5 Joule (B) 2.0 Joule

(C) 2.5 Joule (D) 3.0 Joule

Q5 The figures shows a particle sliding on a frictionless track, which terminates in a straight horizontal section. If the particle starts slipping from the point A, how far away from the track will the particle hit the ground?



(A) $1 \mathrm{~m}$

(B) $2 \mathrm{~m}$

(C) $3 \mathrm{~m}$

(D) $4 \mathrm{~m}$

Q6 A bullet of mass m moving with velocity v block strikes a suspended wooden block of mass M. If the block rises to a height h. The initial velocity of the block will be

(A) $\sqrt{2gh}$

(B) $\frac{\mathrm{M+m}}{\mathrm{m}}\sqrt{\mathrm{gh}}$

(C) $\frac{m}{M+m}2gh$

(D) $\frac{M+m}{M}\sqrt{2gh}$

Q7 Potential energy function along x-axis in a certain force field is given as

$$U(x) = \frac{x^4}{4} - 2x^3 + \frac{11}{2}x^2 - 6x.$$

For the given force field :-

(i) the points of equilibrium are $x = 1, x = 2$ and $x = 3$.

(ii) the point $\mathrm{x} = 2$ is a point of unstable equilibrium.

(iii)the points $x = 1$ and $x = 3$ are points of stable equilibrium.

(iv)there exists no point of neutral equilibrium.

The correct option is :-

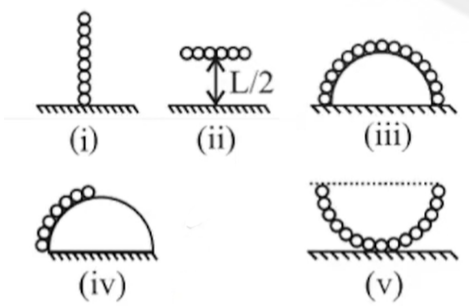
(A) (i), (ii), (iv)

(B) (i), (ii), (iii), (iv)

(C) (iii), (iv)

(D) (ii), (iii)

Q8 A chain of length L and mass M is arranged as shown in following four cases. The correct decreasing order of potential energy (assumed zero at horizontal surface) is:



(A) i > ii > iii > iv > v

(B) i = ii > iii > iv > v

(C) i = ii > iv > iii > v

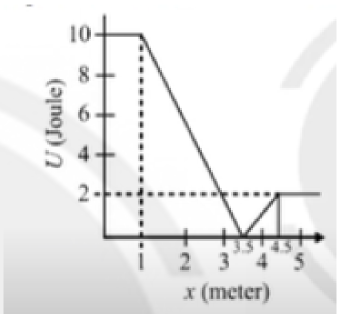
(D) i = ii > iv > v > iii

Q9 A simple pendulum has a string of length l and bob of mass m. When the bob is at its lowest position, it is given the minimum horizontal speed necessary for it to move in a circular path about the point of suspension. The tension in the string at the lowest position of the bob is :

(A) $3mg$ (B) $4mg$

(C) $5mg$ (D) $6mg$

Q10 A body with mass $2\ \mathrm{kg}$ moves in one direction in the presence of a force which is described by the potential energy graph. If the body is released from rest at $x = 2\ \mathrm{m}$, then its speed when it crosses $x = 5\ \mathrm{m}$ is



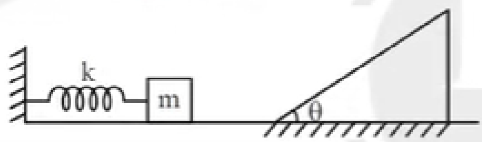
(A) Zero

(B) $1\ \mathrm{ms}^{-1}$

(C) $2\ \mathrm{ms}^{-1}$

(D) $3\ \mathrm{ms}^{-1}$

Q11 In the given situation, the spring is compressed by the block by an amount x. If the block is released then it will rise to a height h on the inclined plane. Value of h will be:



(A) $\frac{mg}{x}$

(B) $\frac{kx^2}{mg}$

(C) $\frac{kx^2}{2mg}$

(D) $\frac{mg}{2kx^2}$

Q12 The blades of a windmill sweep out a circle of area A. If the wind flows at a velocity v

perpendicular to the circle, then the mass of the air of density ρ passing through it in time t is

(A) $Av\rho t$ (B) $2Av\rho t$

(C) $Av^2\rho t$ (D) $\frac{1}{2}Av\rho t$

Q13 A particle is kept at rest at the top of a sphere of diameter $42\ \mathrm{m}$. When disturbed slightly, it slides down. At what height h from the bottom, the particle will leave the sphere

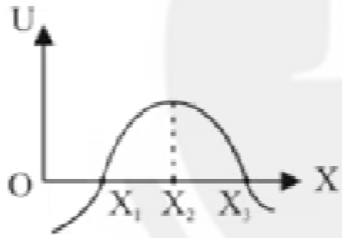
(A) $14\ \mathrm{m}$

(B) $28\ \mathrm{m}$

(C) $35\ \mathrm{m}$

(D) $7\ \mathrm{m}$

Q14 In the figure shown the potential energy (U) of a particle is plotted against its position x from origin. Then which of the following statement is correct. A particle at:



(A) x_1 is in stable equilibrium

(B) x_2 is in stable equilibrium

(C) x_3 is in stable equilibrium

(D) None of these

Q15 A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement x is proportional to-

(A) x^2

(B) e^{x}

(C) x

(D) $\log_e \mathrm{x}$

Q16 A pump is used to deliver water at a certain rate from a given pipe. To obtain n times water from the same pipe in the same time, by what factor, the force of the motor should be increased?

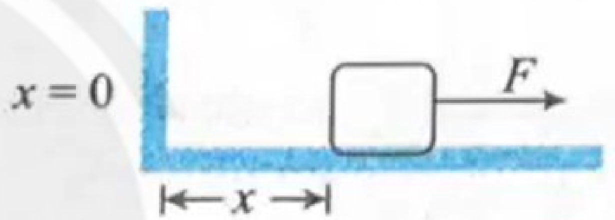
(A) n times

(B) n^2 times

(C) n^3 times

(D) $\frac{1}{n}$ times

Q17 The block of mass m initially at $x=0$ is acted upon by a horizontal force $F=a-bx^2$ (where $a>\mu mg$), as shown in the figure. The co-efficient of friction between the surfaces of contact is μ. The net work done on the block is zero, if the block travels a distance of



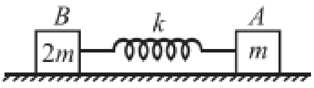
(A) $x=\sqrt{\frac{3(a-\mu mg)}{b}}$

(B) $x=\sqrt{\frac{3(a+\mu mg)}{b}}$

(C) $x=\sqrt{\frac{2(a-\mu mg)}{b}}$

(D) $x=\sqrt{\frac{2(a+\mu mg)}{b}}$

Q18 Two blocks A and B of mass m and $2m$ respectively are connected by a massless spring of spring constant k. This system lies over a smooth horizontal surface. At $t=0$, the block A has velocity u towards right as shown while the speed of block B is zero, and the length of spring is equal to its natural length at that instant.



smooth horizontal surface

	Column-I		Column-II
(A)	The velocity of block A	(p)	can never be zero
(B)	The velocity of block B	(q)	may be zero at certain instant of time
(C)	The kinetic energy of system of two blocks	(r)	is minimum at maximum compression of spring
(D)	The potential energy of spring	(s)	is maximum at maximum compression of spring

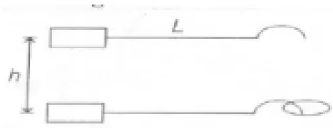
(A) $\mathrm{A} \rightarrow \mathrm{q}; \mathrm{B} \rightarrow \mathrm{q}; \mathrm{C} \rightarrow \mathrm{p}, \mathrm{r}; \mathrm{D} \rightarrow \mathrm{q}, \mathrm{s}$

(B) $\mathrm{A} \rightarrow \mathrm{q}, \mathrm{s}; \mathrm{B} \rightarrow \mathrm{q}; \mathrm{C} \rightarrow \mathrm{p}, \mathrm{q}; \mathrm{D} \rightarrow \mathrm{q}, \mathrm{s}$

(C) $\mathrm{A} \rightarrow \mathrm{q}; \mathrm{B} \rightarrow \mathrm{q}; \mathrm{C} \rightarrow \mathrm{q}, \mathrm{p}; \mathrm{D} \rightarrow \mathrm{q}, \mathrm{r}$

(D) None of these

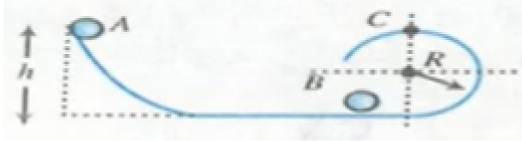
Q19 The figure shows a light rod of length L rigidly attached to a small heavy block at one end and a hook at the other end. The system is released from rest with the rod in a horizontal position. There is a fixed smooth ring at a depth h below the initial position of the hook and the hook gets into the ring as it reaches there. What should be the minimum value of h, so that the block moves in a complete circle about the ring?



(A) $h = L$ (B) $h = 2L$

(C) $h = 3L$ (D) $h = 4L$

Q20 Ball A of mass m, after sliding from an inclined plane, strikes elastically another ball B of same mass at rest. Find the minimum height h so that ball B just completes the circular motion of the surface at C. (All surfaces are smooth.)



(A) $n = \frac{5}{2}R$

(B) $h = 2R$

(C) $h = \frac{2}{5}R$

(D) $h = 3R$

Q21 In the List-I below, four different paths of a particle are given as functions of time. In these functions, α and β are positive constants of appropriate dimensions and $\alpha \neq \beta$. In each case, the force acting on the particle is either zero or conservative. In List-II, five physical quantities of the particle are mentioned: $\vec{\mathrm{p}}$ is the linear momentum, $\vec{\mathrm{L}}$ is the angular momentum about the origin, K is the kinetic energy, U is the potential energy and E is the total energy. Match each path in List-I with those quantities in List-II, which are conserved for that path.

List-I		List-II
P. $\vec{r}(t)=\alpha t\,\hat{i}+\beta t\,\hat{j}$	**1.**	$\vec{p}$
Q. $\vec{r}(t)=\alpha\cos\omega t\,\hat{i}+\beta\sin\omega t\,\hat{j}$	**2.**	$\vec{L}$
R. $\vec{r}(t)=\alpha(\cos\omega t\,\hat{i}+\sin\omega t\,\hat{j})$	**3.**	K
S. $\vec{r}(t)=\alpha t+\frac{\beta}{2}t^2\,\hat{j}$	**4.**	U
	5.	E

(A) $\mathrm{P} \rightarrow 1,2,3,4,5; \mathrm{Q} \rightarrow 2,5; \mathrm{R} \rightarrow 2,3,4,5; \mathrm{S} \rightarrow 5$

(B) $\mathrm{P} \rightarrow 1,2,3,4,5; \mathrm{Q} \rightarrow 3,5; \mathrm{R} \rightarrow 2,3,4,5; \mathrm{S} \rightarrow 2,5$

(C) $\mathrm{P} \rightarrow 2,3,4; \mathrm{Q} \rightarrow 5; \mathrm{R} \rightarrow 1,2,4; \mathrm{S} \rightarrow 2,5$

(D) $\mathrm{P} \rightarrow 1,2,3,5; \mathrm{Q} \rightarrow 2,5; \mathrm{R} \rightarrow 2,3,4,5; \mathrm{S} \rightarrow 2,5$

Q22 A smooth chain AB of mass m rests against a surface in the form of a quarter of a circle of radius R. If it is released from rest, the velocity of the chain after it comes over the horizontal part of the surface is

(A) $\sqrt{2gR}$

(B) $\sqrt{gR}$

(C) $\sqrt{2gR\left(1-\frac{2}{\pi}\right)}$

(D) $\sqrt{2\,gR(2-\pi)}$

Q23 A wind-powered generator converts wind energy into electric energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v, the electrical power output will be proportional to

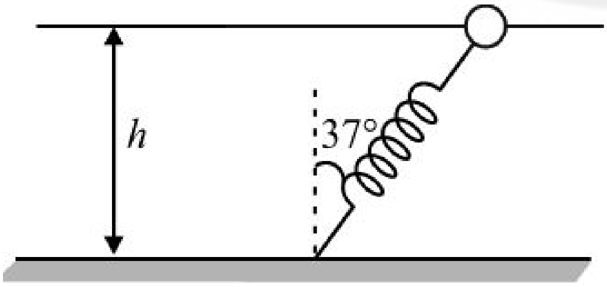
(A) v

(B) v^2

(C) v^3

(D) v^4

Q24 One end of a spring of natural length *h* and spring constant *k* is fixed at the ground and the other is fitted with a smooth ring of mass *m* which is allowed to slide on a horizontal rod fixed at a height *h*. Initially the spring makes an angle of 37° with the vertical when the system is released from rest. The speed of the ring when the spring becomes vertical is



(A) $\sqrt{\frac{k}{m}}$

(B) $\frac{h}{2}\sqrt{\frac{k}{m}}$

(C) $\frac{h}{3}\sqrt{\frac{k}{m}}$

(D) $\frac{h}{4}\sqrt{\frac{k}{m}}$

Q25 If a particle is moving on straight line and a constant instantaneous power is supplying on the particle then find displacement of particle as a function of time.

(A) $x=\left[\frac{1}{3}\left(\frac{3c}{m}\right)^{1/3}t\right]^{3/2}$

(B) $x=\left[\frac{5}{3}\left(\frac{3c}{m}\right)^{1/3}t\right]^{3/2}$

(C) $x=\left[\frac{2}{3}\left(\frac{3c}{m}\right)^{1/3}t\right]^{3/2}$

(D) $x=\left[\frac{10}{3}\left(\frac{3c}{m}\right)^{1/3}t\right]^{3/2}$

Answer Key

Q1	A
Q2	A
Q3	A
Q4	C
Q5	A
Q6	D
Q7	B
Q8	C
Q9	D
Q10	C
Q11	C
Q12	A
Q13	C
Q14	D
Q15	A
Q16	B
Q17	A
Q18	A
Q19	A
Q20	A
Q21	A
Q22	C
Q23	C
Q24	D
Q25	C